



Electronic I

Lecture 4

Diode Applications : Clippers and Clampers

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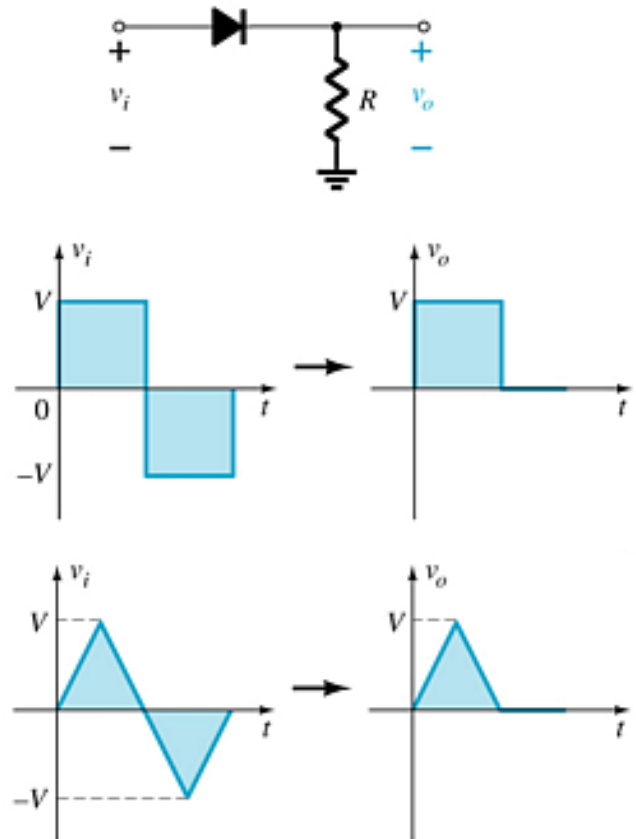
First Year / Spring Semester

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Diode Clippers

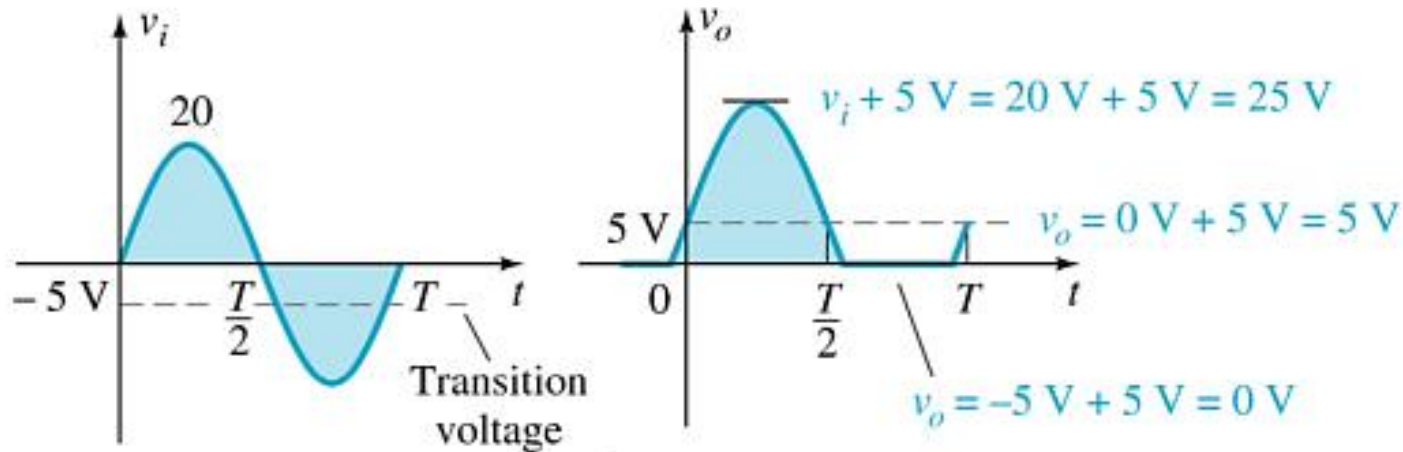
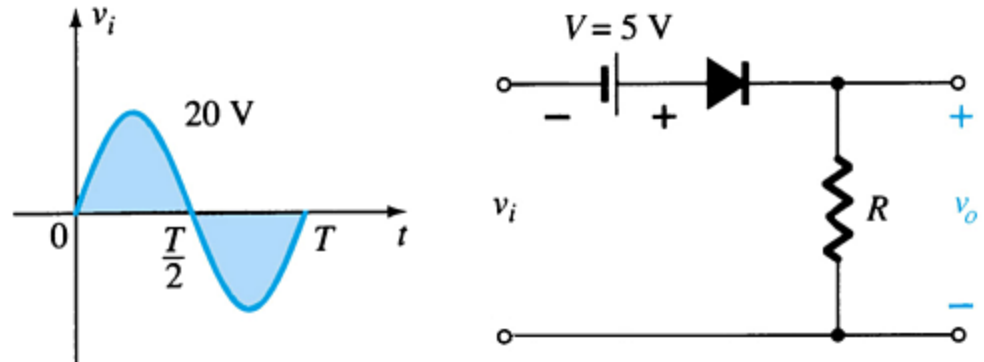
The diode in a **series clipper** “clips” any voltage that does not forward bias it:

- A reverse-biasing polarity
- A forward-biasing polarity less than 0.7 V (for a silicon diode)



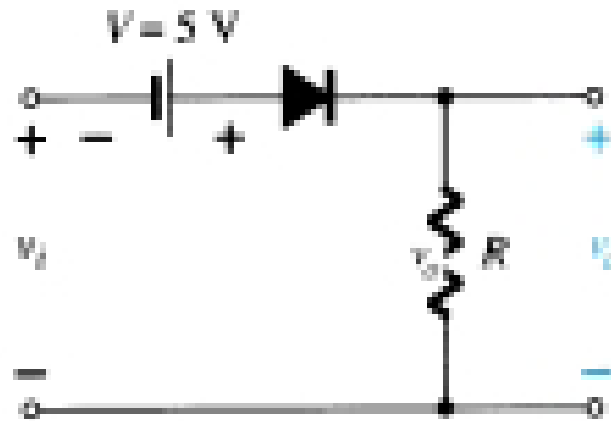
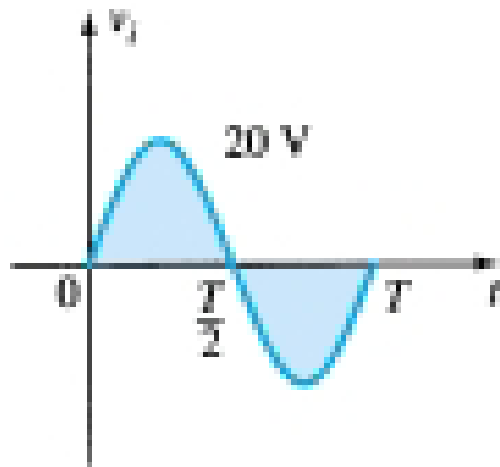
Biased Clippers

Adding a DC source in series with the clipping diode changes the effective forward bias of the diode.

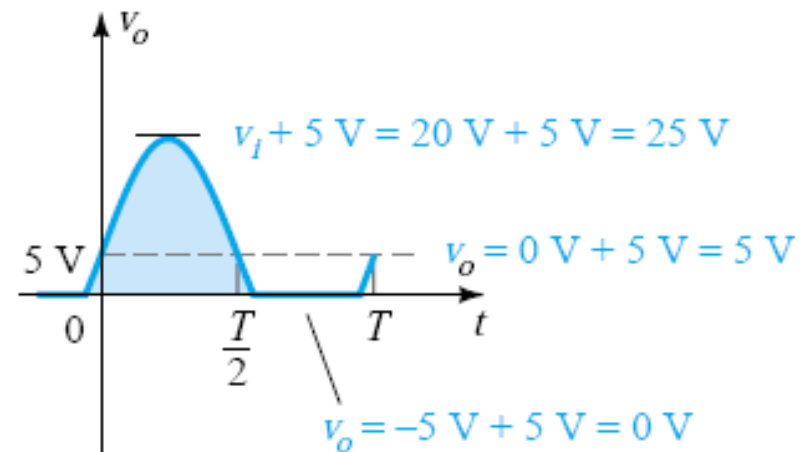
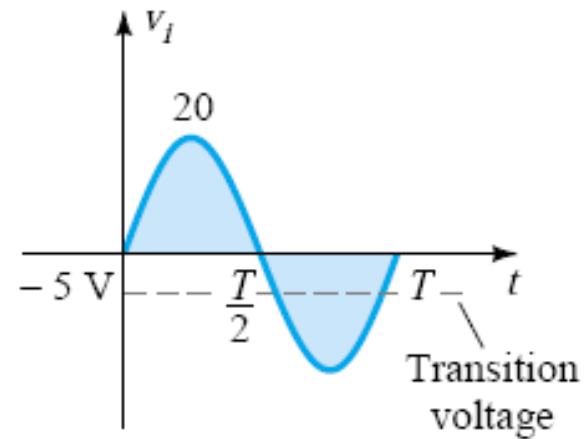
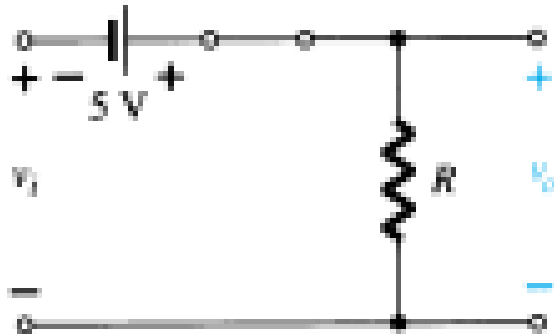


Example

Determine the output waveform for the network

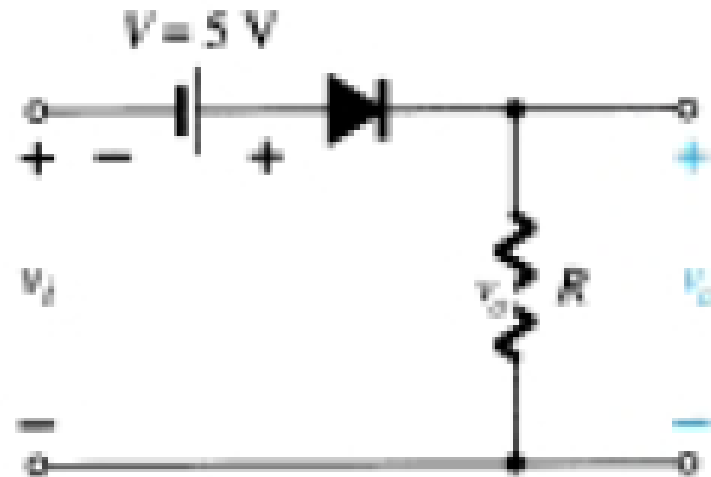
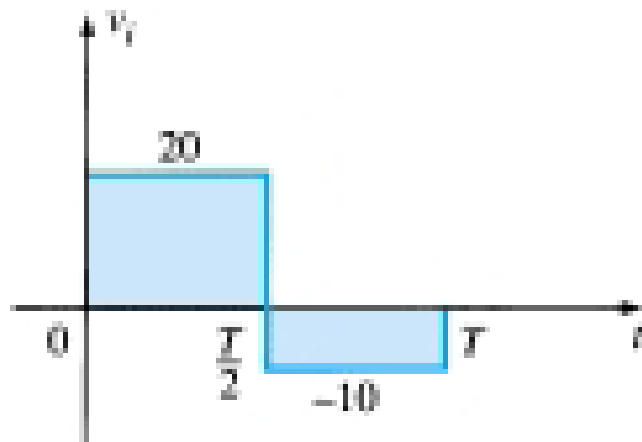


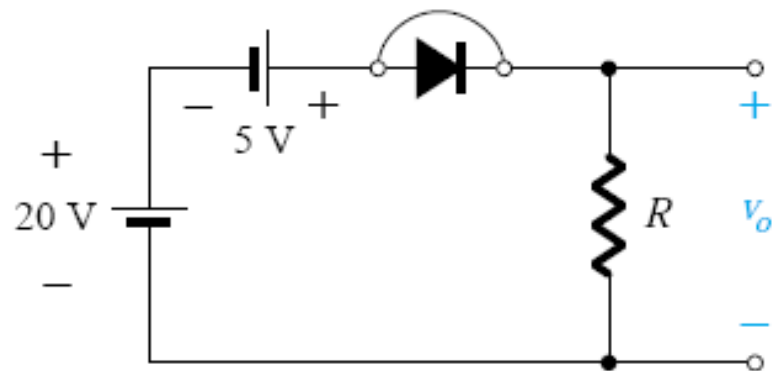
The network will appear as below, and $v_o = v_i + 5V$.



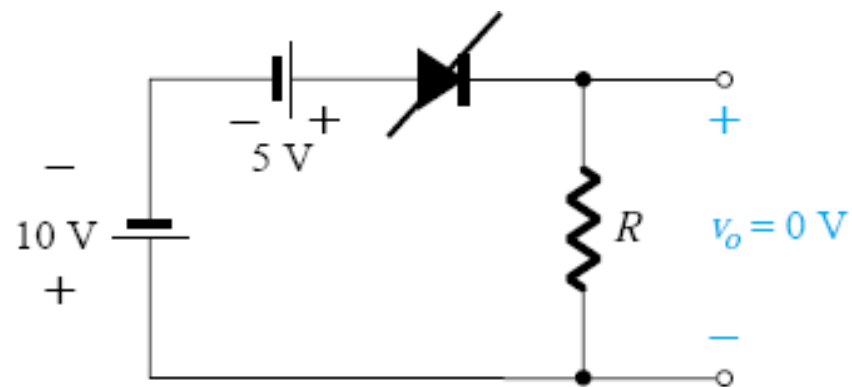
Example

Determine the output waveform for the network

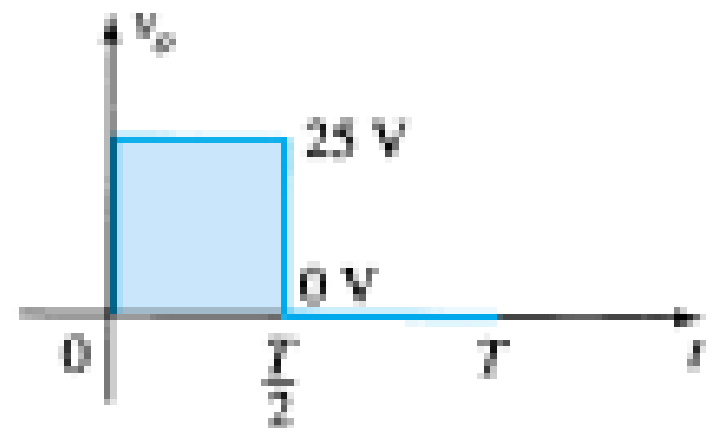




v_o at $v_i = +20$ V.



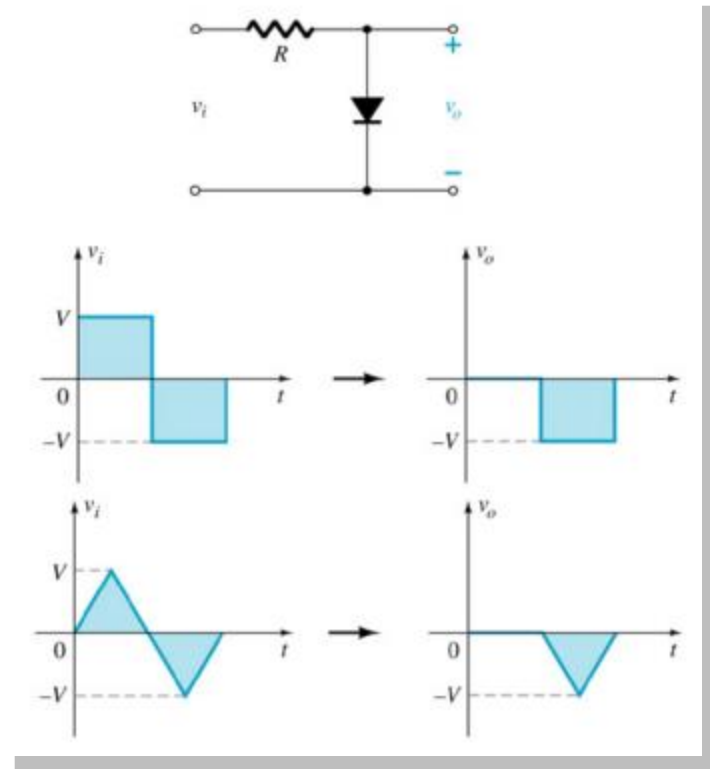
v_o at $v_i = -10$ V.



Parallel Clippers

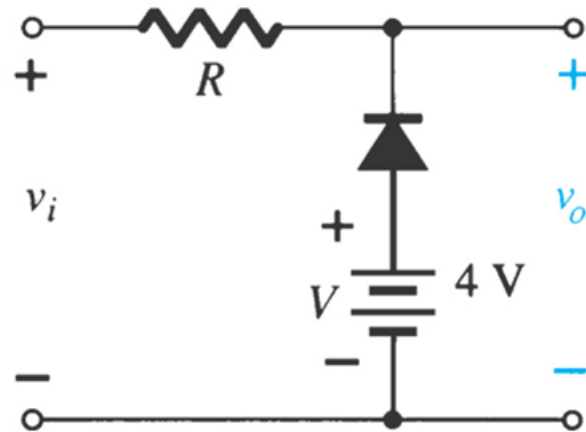
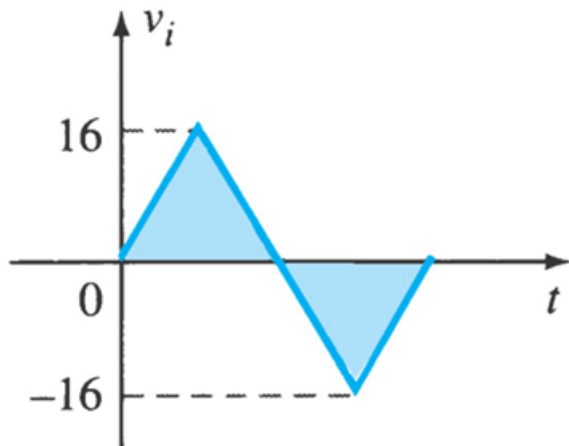
The diode (ideal) in a **parallel clipper** circuit “clips” any voltage that forward bias it.

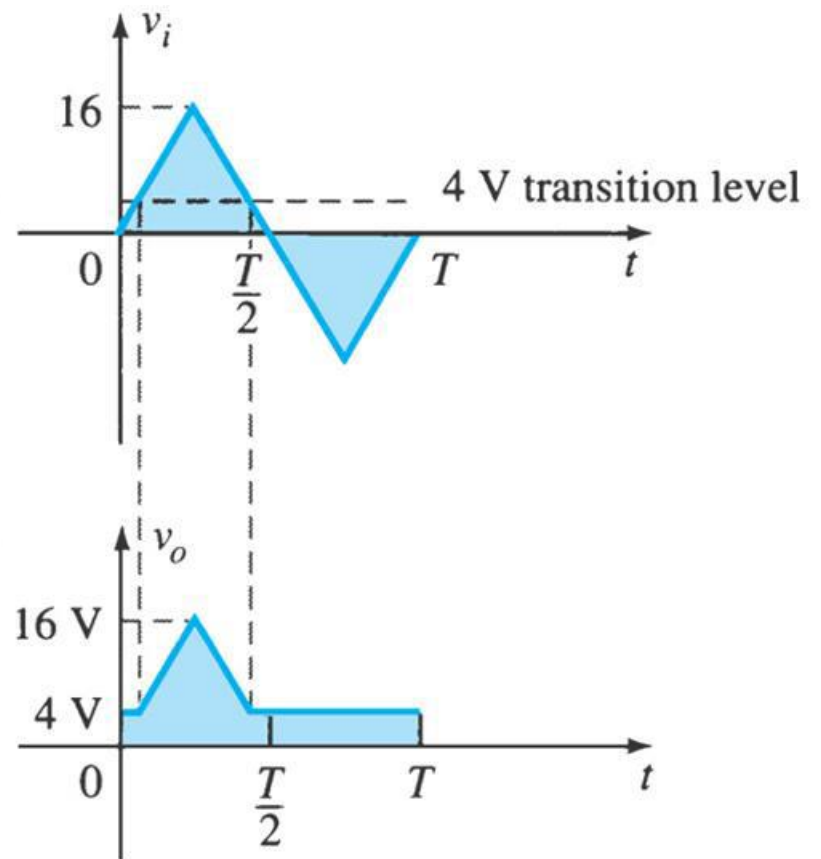
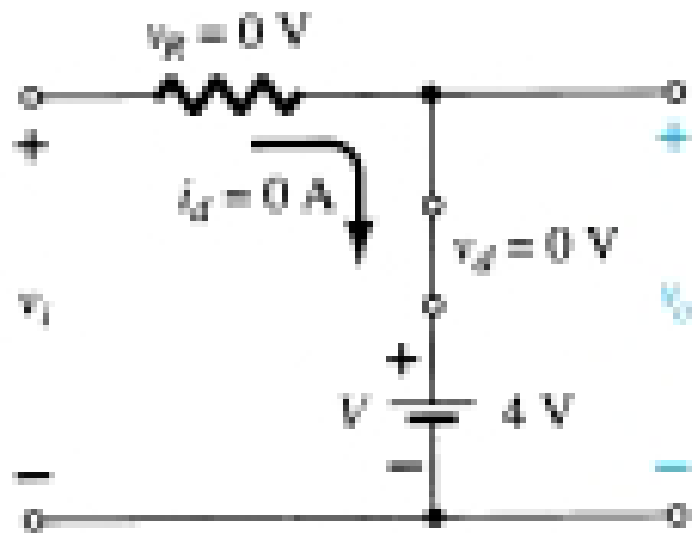
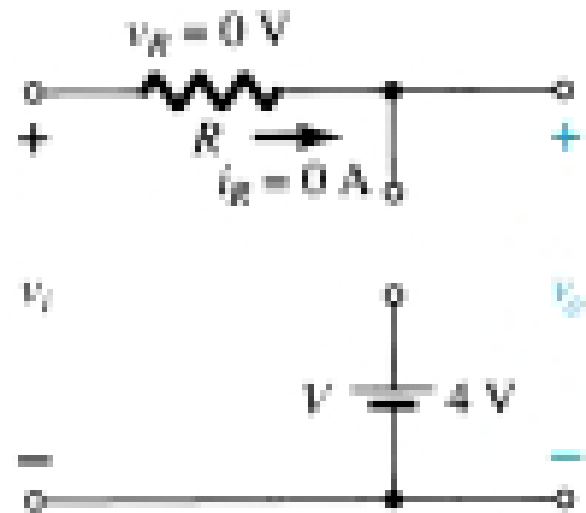
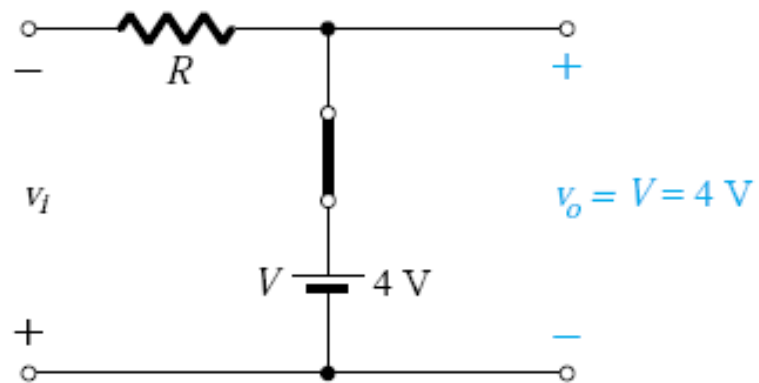
DC biasing can be added in series with the diode to change the clipping level.



Example

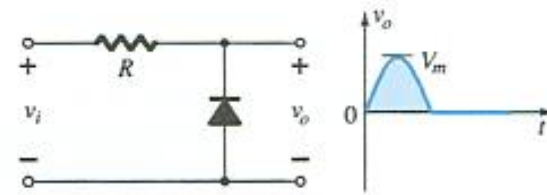
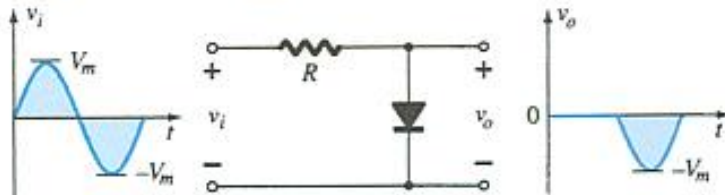
Determine v_o for the following network



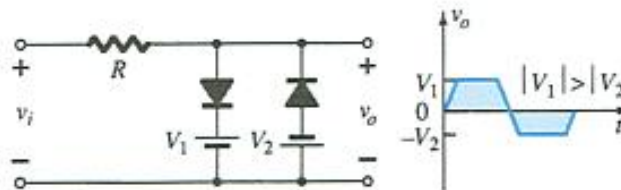
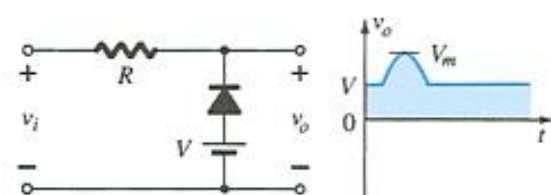
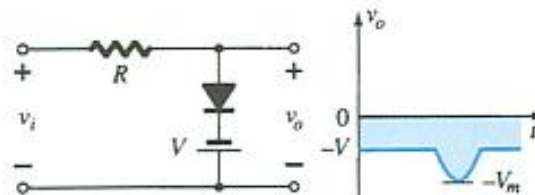
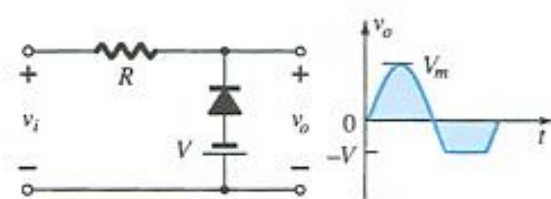
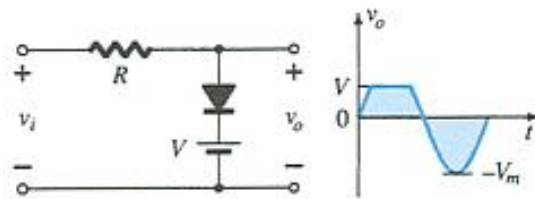


Summary of Clipper Circuits

Simple Parallel Clippers (Ideal Diodes)



Biased Parallel Clippers (Ideal Diodes)

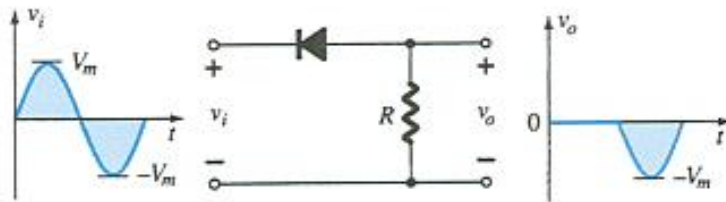


more...

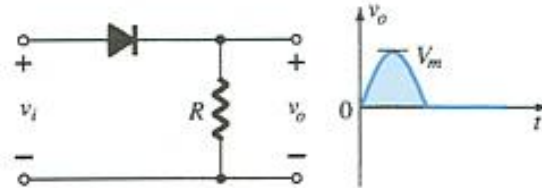
Summary of Clipper Circuits

Simple Series Clippers (Ideal Diodes)

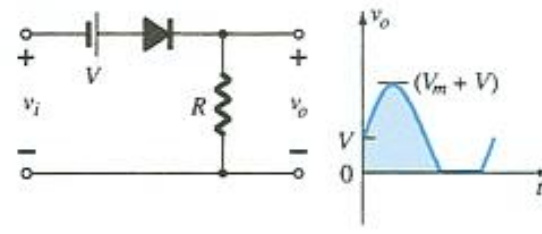
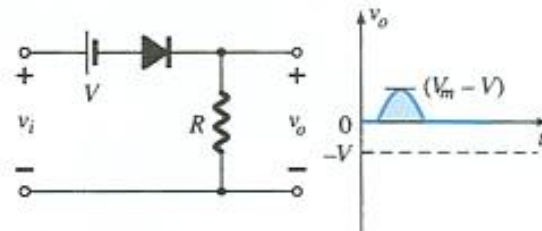
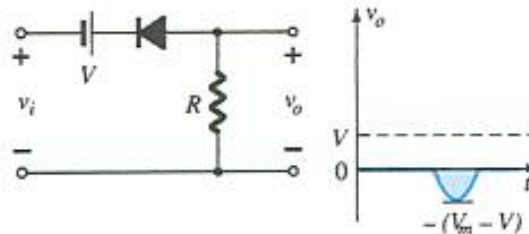
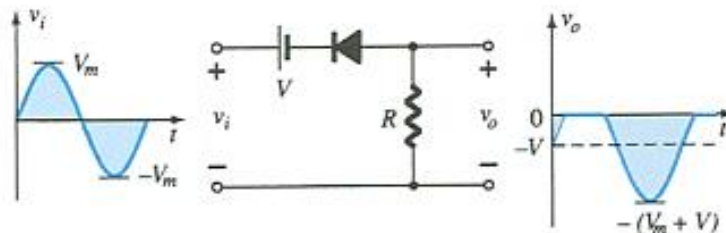
POSITIVE



NEGATIVE

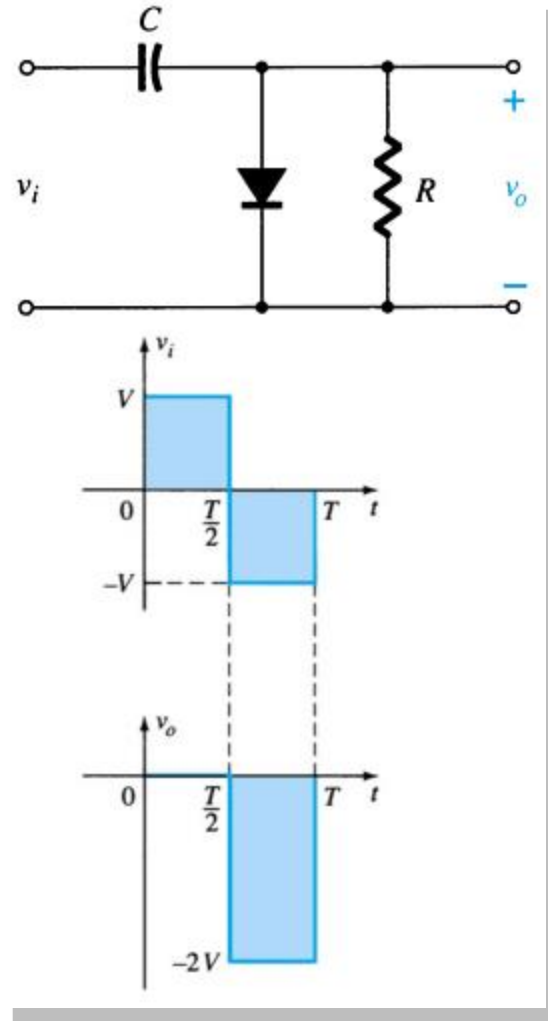


Biased Series Clippers (Ideal Diodes)



Clampers

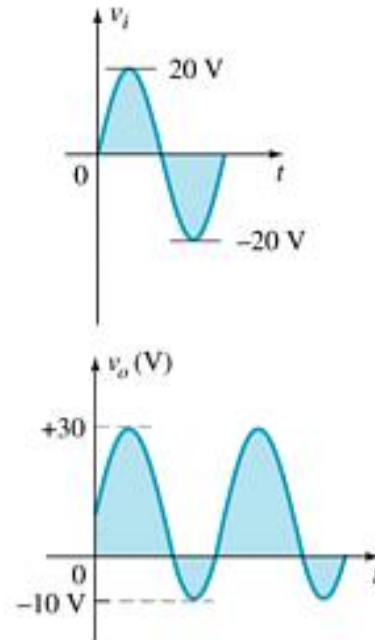
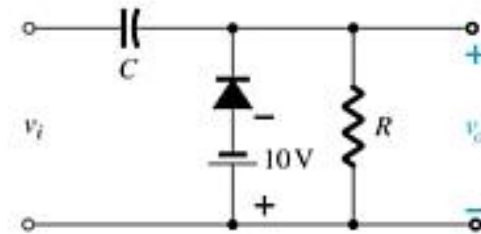
A diode and capacitor can be combined to “clamp” an AC signal to a specific DC level.



Biased Clamper Circuits

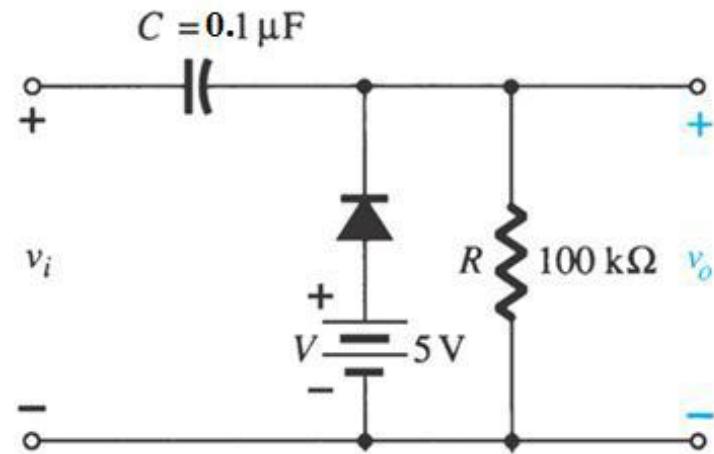
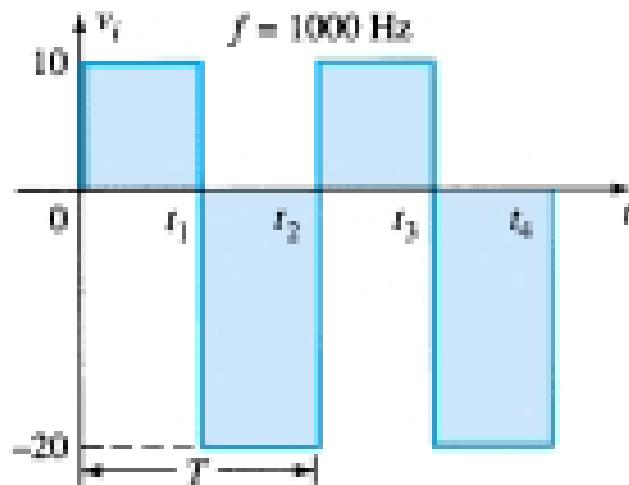
The input signal can be any type of waveform such as sine, square, and triangle waves.

The DC source lets you adjust the DC clamping level.



Example

Determine v_o for the following network for the input indicated.

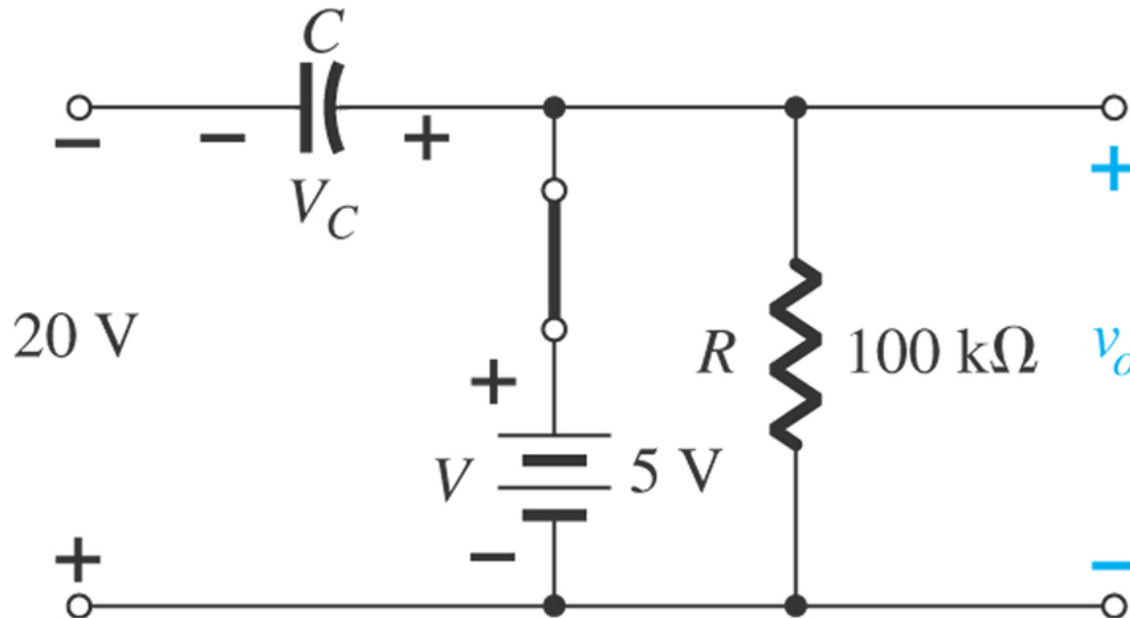


- The analysis will begin with the period $t1 \rightarrow t2$ of the input signal since the diode is in its short-circuit state as recommended by comment 1.
- For this interval the output is across R , but it is also directly across the 5-V battery *the result is $V_o = 5\text{ V}$* for this interval.
- Applying Kirchhoff's voltage law around the input loop will result in

$$-20\text{ V} + V_C - 5\text{ V} = 0$$

$$V_C = 25\text{ V}$$

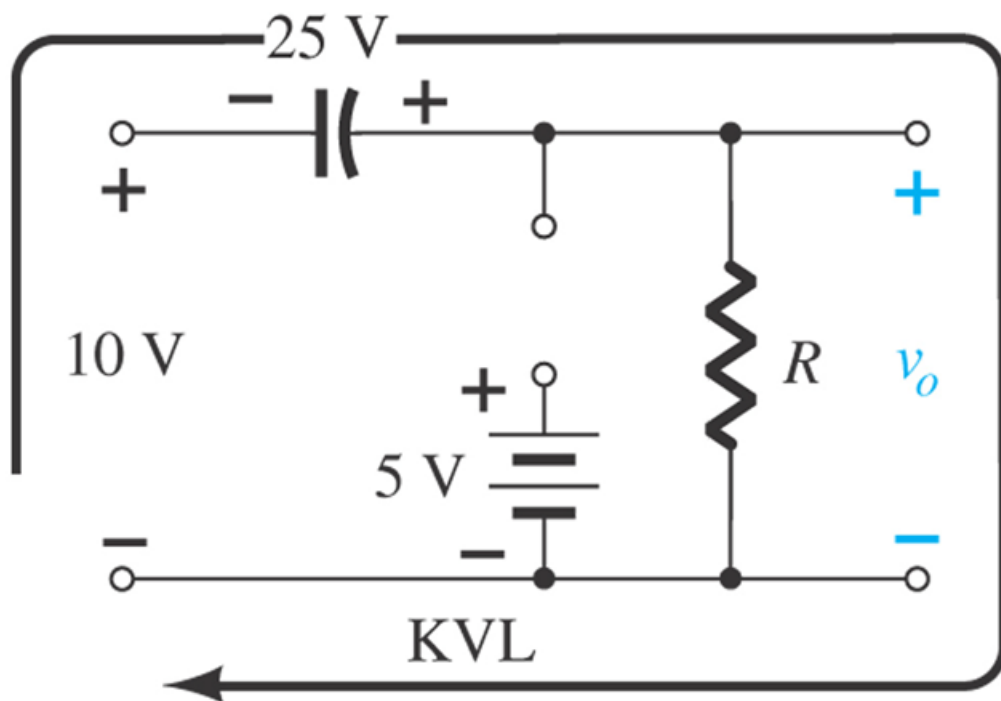
- The capacitor will therefore charge up to 25 V, as stated in comment 2.

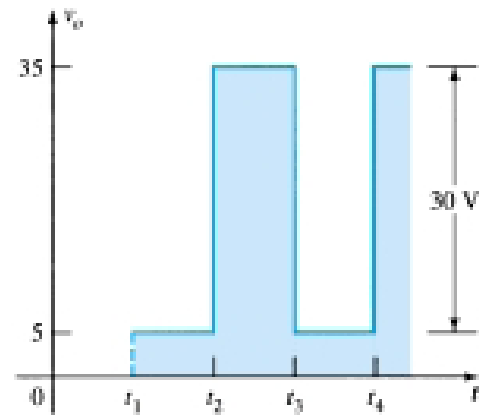
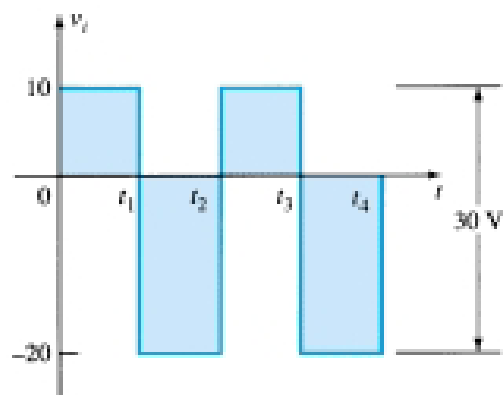
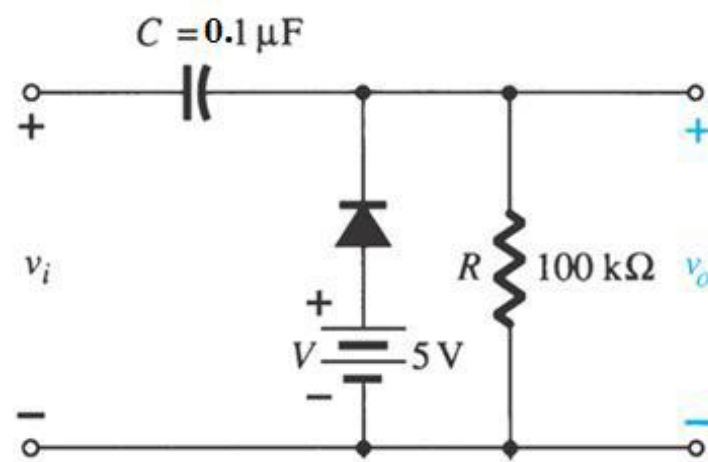


- The open-circuit equivalent for the diode will remove the 5-V battery from having any effect on V_o , and *applying Kirchhoff's voltage law around the outside loop of the network* will result in

$$+10\text{ V} + 25\text{ V} - v_o = 0$$

$$v_o = 35\text{ V}$$





Summary of Clamper Circuits

Clamping Networks

